

COSC 594

LLMs for RL

Dongsheng Ding

Department of Electrical Engineering and Computer Science
University of Tennessee, Knoxville

Reminders

Check Canvas for Project Guideline
Class presentation from 11/3 to 11/19

No Class on Friday 10/31

Outline

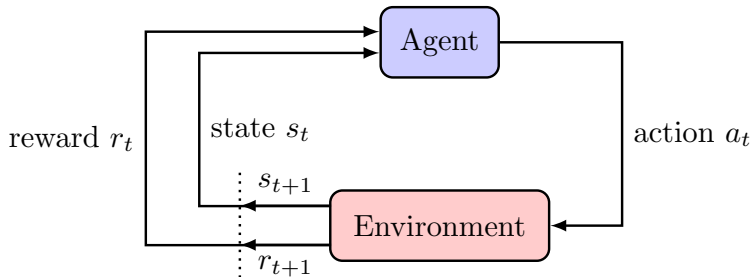
- RECAP
- WORLD MODELS
- LLMS AS WORLD MODELS
- LLMS AS REWARD MODELS

Outline

■ RECAP

RL Framework

■ MARKOV DECISION PROCESSES (MDPs)



Goal $\underset{\pi}{\text{maximize}} \quad \mathbb{E}_{s_0} [V^{\pi}(s_0)]$	$\Leftrightarrow$	Direct policy search $\pi^+ \leftarrow \pi + \nabla_{\pi} V^{\pi}$
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π – a policy that maps states to actions

$V^{\pi}(s_0) := \mathbb{E} [\text{sum of all rewards received}]$

Policy optimization

Policy: $\pi : S \text{ (states)} \rightarrow \Delta_A \text{ (action distribution)}$

Value function: $V^\pi(\rho) := \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 \sim \rho \right]$

$P(\cdot \mid s, a) : S \times A \rightarrow \Delta_A$ – transition kernel

$s_0, a_0, r_0, s_1, a_1, \dots$ – a trajectory τ generated by a policy π

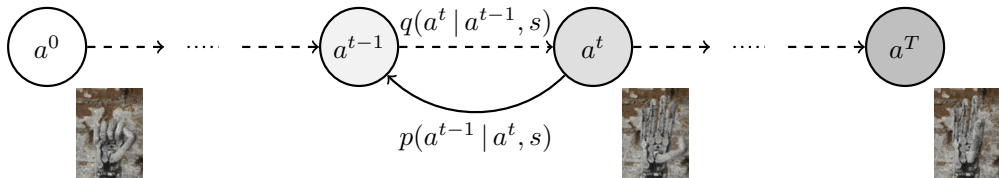
■ NONCONVEX OPTIMIZATION

$$\underset{\pi}{\text{maximize}} \quad V^\pi(\rho)$$

★ Search the decision in [a function space](#)

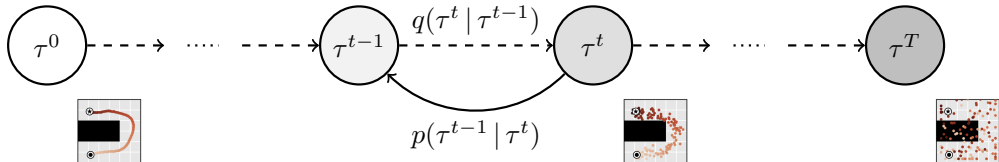
Generative models for RL

■ DIFFUSION MODELS AS POLICIES



data point a^0 – action

■ DIFFUSION MODELS AS TRAJECTORY GENERATORS



data point τ^0 – trajectory

Outline

- RECAP

- WORLD MODELS

World models

How the **world state** changes after taking an action

Transition kernel: $P(\cdot \mid \textcolor{red}{s}, a) : S \times A \rightarrow \Delta_A$

Can we use LLMs as world models to predict the world state changes?

What is a world model?

■ REPRESENTATION OF ENVIRONMENT

“**Representations** which preserve the **causal structure** of the environment as far as is necessitated by the tasks an agent needs to perform” (source: arXiv:2412.11867)

“Mental **representation** of the environment, which enables humans to **simulate** actions and their effects on the **world's state** for deliberate planning for complex tasks of motor control, imagery, inference, and decision making” (source: arXiv:2305.14992)

Representation w/ task-associated structure

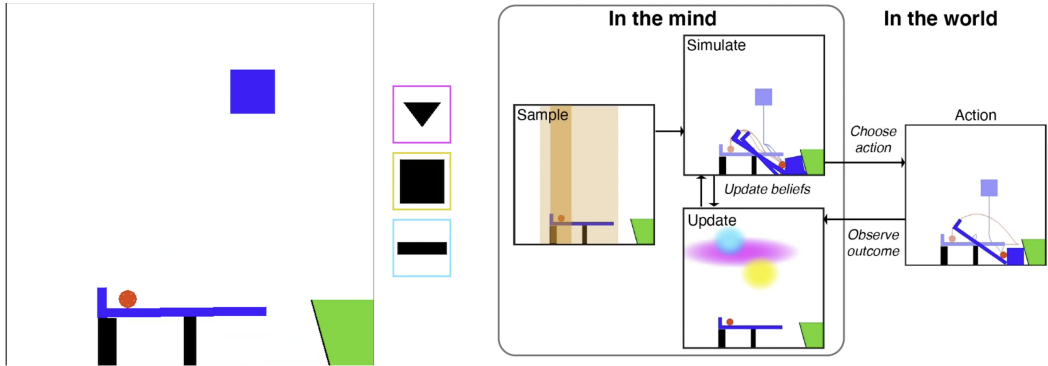
World models in human: physics



Humans – even infants – can reason about and interact with liquids

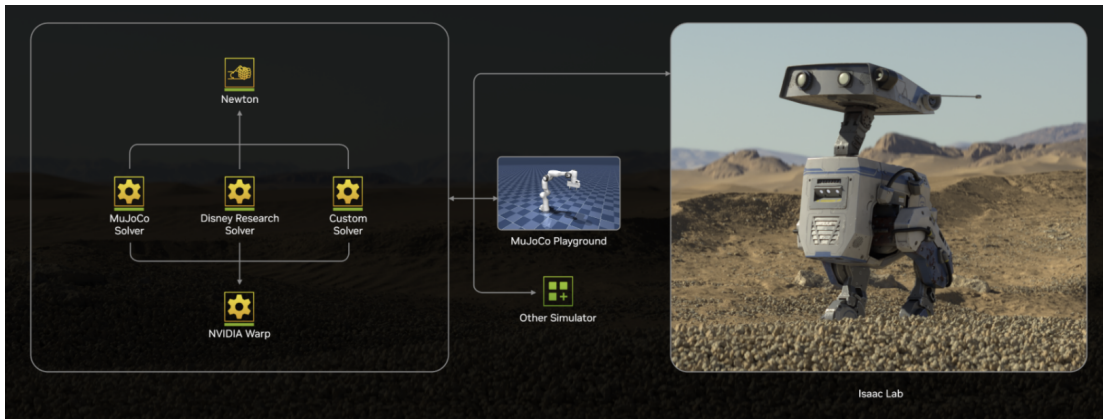
image source: <https://escholarship.org/uc/item/6xc533j2>

World models in human: tool use



**Humans imagine the effects of hypothesized action via
Sample, Simulate, Update**

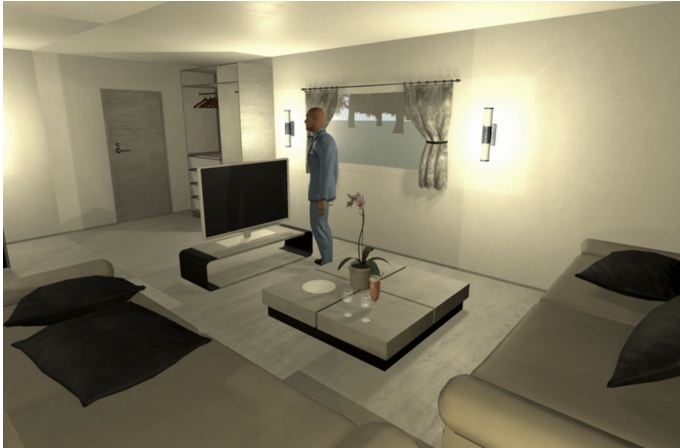
World models in simulation: physics engines



Newton, Physics Engine for Robotics Simulation

Accurate virtual representations of robot behavior and interactions within realistic environments

World models in simulation: embodied simulators



VirtualHome, Multi-Agent Household Simulator

Simulate activities in a household

World models in simulation: video prediction models



GAIA-2, Synthetic Scenario Generation for Real-world Safety

Generate realistic, diverse, and precisely controlled scenarios

World models are typically domain specific

Language encodes or structures nearly all human knowledge

LLMs $\approx$ Language $\rightarrow$ LLMs as world models

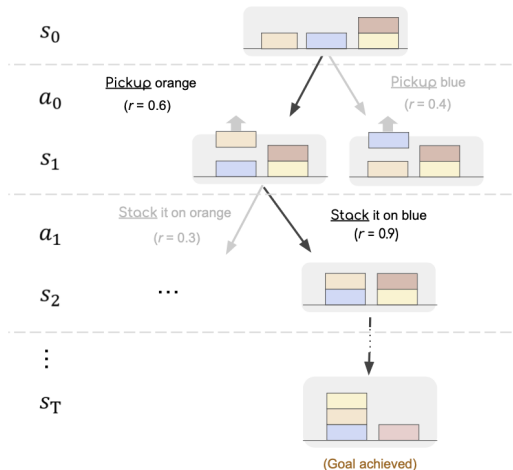
Outline

- RECAP
- WORLD MODELS
- LLMS AS WORLD MODELS

Blocksworld example

Initial State: The orange block is on the table, the blue block is on the table, and the red block...

Goal: The orange block is on the blue block, and the yellow block is on the orange block.



World model-based MDP

■ STATE

s_t = configuration of blocks in language

■ ACTION

a_t = a behavior of moving a block, e.g., pickup orange

■ NEXT STATE

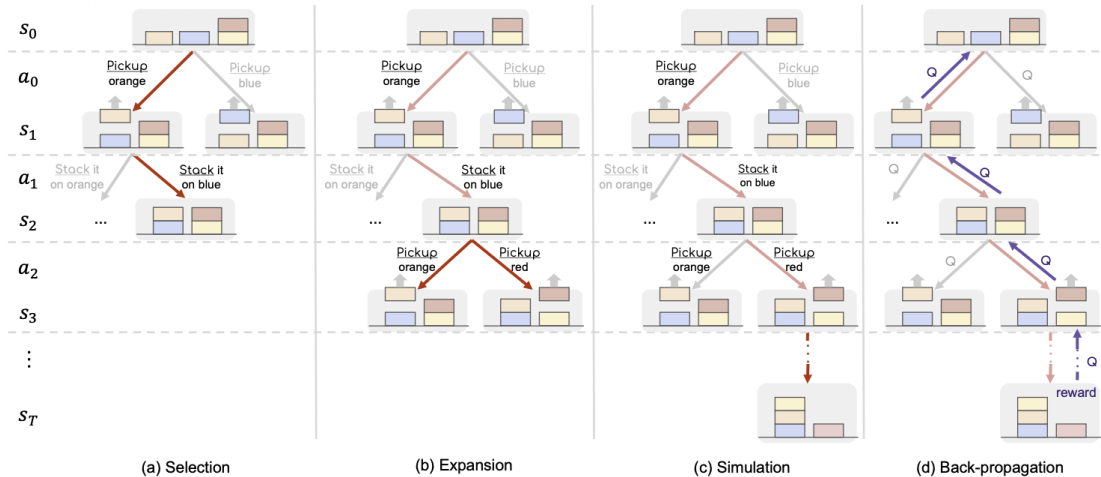
s_{t+1} = new configuration of blocks, given (s_t, a_t)

LLM as World Model

■ REWARD

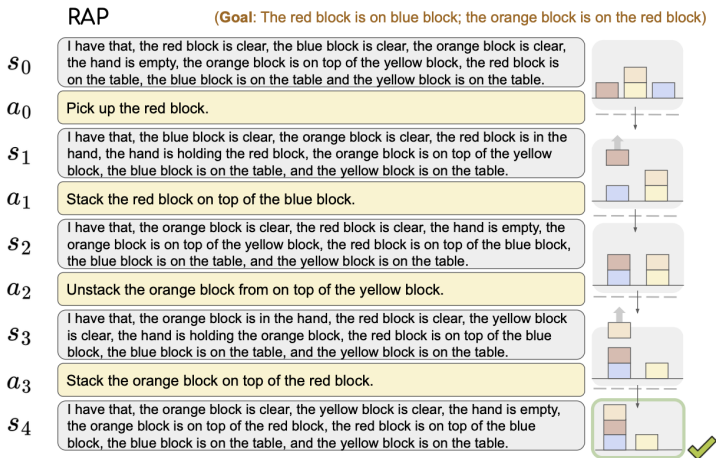
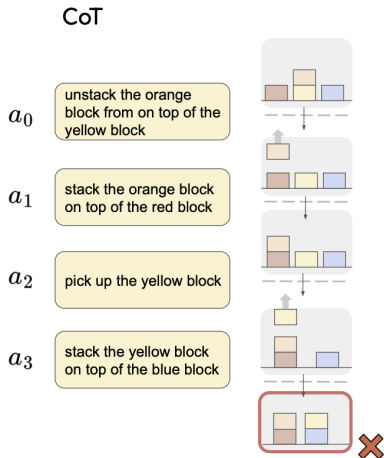
$r(s_t, a_t)$ = e.g., LLM's preference and self-evaluation

Monte Carlo tree search planning



Expected future reward $Q(s, a)$

Monte Carlo tree search planning



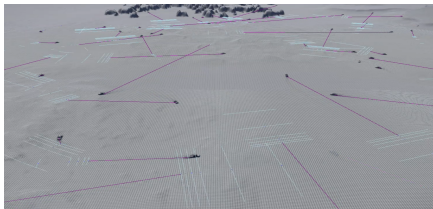
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Required behavior is complex

$$\begin{aligned} r(s, a) = & w_0 b_1(s, a) && \text{path following} \\ & + w_1 b_2(s, a) && \text{speed limit} \\ & + w_2 b_3(s, a) && \text{energy budget} \\ & + w_3 b_4(s, a) && \text{acceleration limit} \\ & \vdots && \simeq 400 \text{ lines of code, but hopefully work!} \end{aligned}$$

Unforeseen behavior in navigation



UBISOFT '22

Reward hacking

■ ENVIRONMENT OR GOAL MISSPECIFIED



image source: <https://openai.com/index/faulty-reward-functions/>

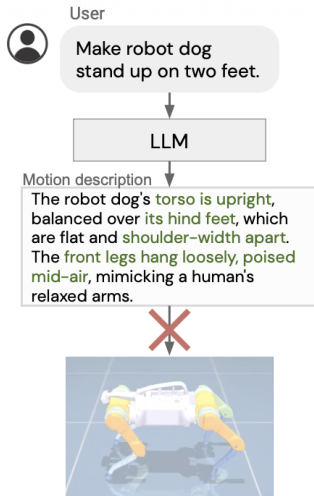
Reward design is expert-dependent and subjective

Language encodes rich knowledge about robot motions

LLMs $\approx$ Language $\rightarrow$ LLMs as reward models

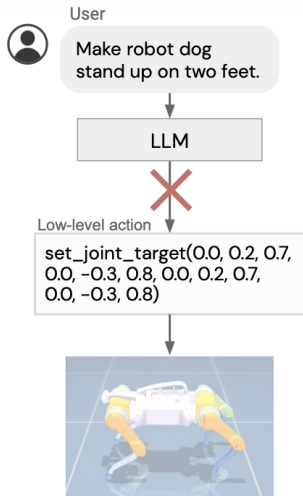
Gap between language and robot actions (I)

■ FROM LANGUAGE AND LOW-LEVEL ROBOT ACTIONS



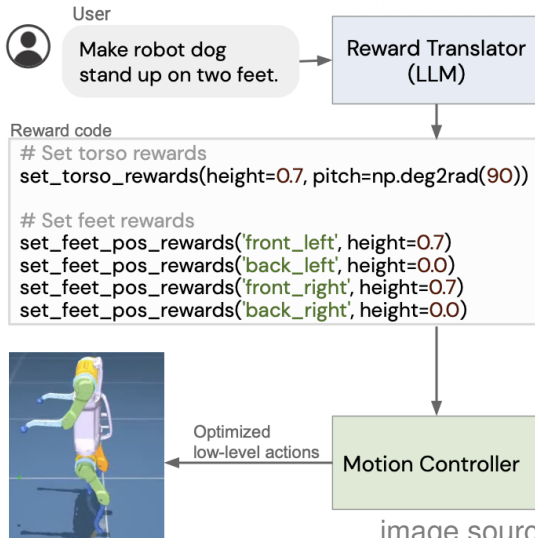
Gap between language and robot actions (II)

■ LACK OF UNDERSTANDING OF LOW-LEVEL ROBOT ACTIONS



LLMs as reward models

REWARD TRANSLATOR



Reward interface

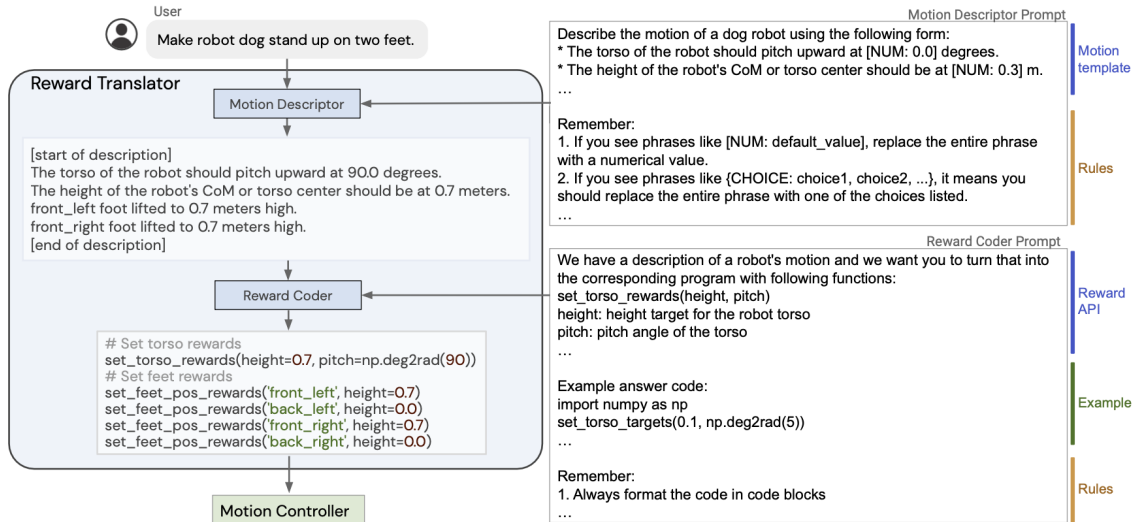
■ INSTRUCTION-BASED REWARD

$$r(s, a) = - \sum_{i=1}^m w_i \underbrace{f_i(z_i(s, a, \psi_i))}_{\text{task-specific}}$$

- ★ loss function f_i takes minimum at 0, e.g., $f_i(\cdot) = \|\cdot\|$
- ★ residual term z_i measures error
- ★ reward parameter ψ_i encodes task requirement

e.g., robot height s reaches a desired height ψ_i : $z_i(s, \psi_i) = s - \psi_i$

Reward translator



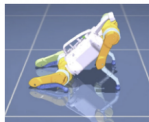
Motion controller

■ OPTIMAL CONTROL

$$\underset{a_{1:H}}{\text{maximize}} \quad \sum_{t=1}^H r(s_t, a_t)$$

- ★ reinforcement learning
- ★ trajectory optimization
- ★ model predictive control

Experiment



Sit down



Lift a paw



Roll over

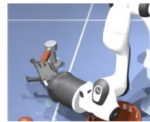


Biped stand



Face sunset

(a) Quadruped robot



Turn on faucet



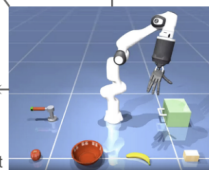
Open drawer



Flip the bowl



Keep banana upright



Lift the apple

(b) Dexterous manipulator robot

Tasks for quadruped robot

Task	Instructions	Expected Behavior
Facing sunrise	It's early in the morning, make the robot head towards the sun.	Robot face towards East.
Facing sunset	It's late in the afternoon, make the robot head towards the sunset.	Robot face towards West.
Sit down	Sit down low to ground with torso flat.	Robot's CoM drops lower and remain flat.
Roll Over	I want the robot to roll by 180 degrees.	Robot's belly faces up.
Spin	Spin fast.	Robot reach a fast turning speed.
Lift one paw	I want the robot to lift its front right paw in the air.	The front right paw of the robot lifts up in the air.
Lift paw higher	I want the robot to lift its front right paw in the air. Lift it even higher.	The robot lifts its front right paw higher than before.
Spin with lifted paws	Lift front left paw. Good, now lift diagonal paw as well. Good, in addition I want the robot to spin fast.	Robot lifts front left and rear right paws while spin fast.
Stand up on two feet	Make the robot stand upright on two back feet like a human.	Robot stands on two back feet and keep balance.

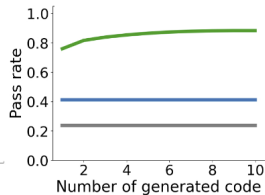
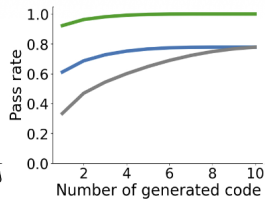
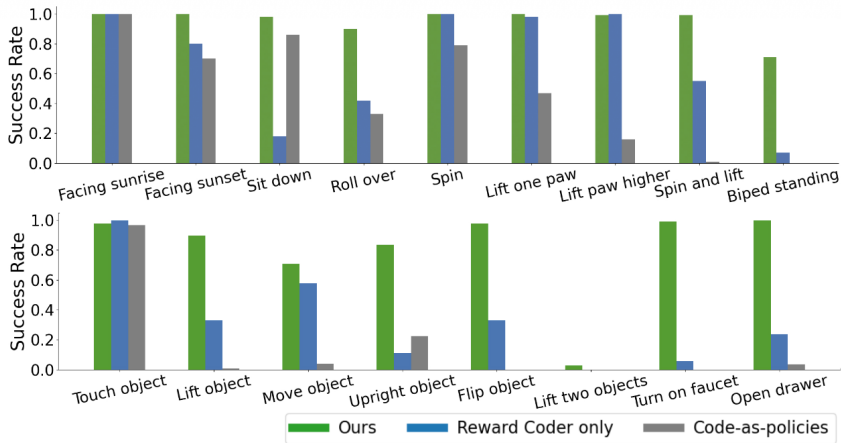
Table 1: List of tasks used in evaluation for the quadruped robot.

Tasks for dexterous manipulation

Task	Instructions	Expected Behavior
Touch object	Touch the {object}	Robot fingers in contact with the object.
Lift object	Lift the {object} to 0.5 m	The object needs to stay above 0.4 m for 1s.
Move object	Move the {object_a} to {object_b}	The distance between object needs to be smaller than 0.1 m.
Upright object	Place the {object} upright	The z axis of the object needs to be parallel to x-y plane.
Flip object	Flip the {object}	The local up vector of the object should be pointing downward.
Lift two objects	Lift the {object_a} and {object_b} at the same time.	Both objects need to stay above 0.4 m for 1 s.
Turn on the faucet	Turn on the faucet.	The valve of the faucet needs to be turned 90 degrees.
Open the drawer	Open the drawer.	The drawer needs to be pulled fully open.

Table 2: List of tasks used in evaluation for the dexterous manipulation.

Success rate



Limitations of LLMs

■ LLMS LACK WORLD AND AGENT KNOWLEDGE



Emily Zhou
Female, 29 years old
Software Engineer (Web Crawler Team)



Good morning! I'd like to schedule a meeting with you and Liu. When are you available?
9:43 AM



Hi there, I'm available for a meeting on Wednesday or Thursday. What time works best for you on those days?
10:03 AM



Great! I'm also available on Wednesday afternoons or Thursday mornings. Let me check Liu Qiang's schedule and confirm.
10:04 AM



Sounds good. Once you've checked with Liu, let me know the final time for Wednesday or Thursday, and I'll block it on my calendar.
10:04 AM



Hello Emily! The final meeting time is 10:30 AM on Thursday.
10:38 AM



Liu Qiang
Male, 36 years old
Quality Assurance Engineer



Hi Liu Qiang, I'd like to schedule a meeting with you and Emily. She's available on Wednesday or Thursday. When are you available?
10:05 AM



Hello! I'm available on Thursday for the meeting with Emily. Please let me know what time works best for both of you.
10:24 AM



Oh good, I think Thursday works for us. Could we aim for a mid-morning slot? Perhaps around 10:30 AM?
10:25 AM



Yup. That works for me. I will send Emily a link myself.
10:37 AM



Thank you. The final meeting time is 10:30 AM on Thursday. Have a great day!
10:38 AM

LLMs fail at 70% of simple office tasks

example source: arXiv:2412.14161

Limitations of LLMs

■ LLMS LACK WORLD AND AGENT KNOWLEDGE

- ★ interacting entities
- ★ social behaviors
- ★ real-world physics

■ MORE WORLD MODELS

- ★ relational reasoning
- ★ social learning
- ★ physics engines

Summary

■ RECAP

■ WORLD MODELS

Representation w/ task-associated structure, LLMs as world models

■ LLMS AS WORLD MODELS

Blocksworld example

■ LLMS AS REWARD MODELS

Robot learning